Digital Image Processing (PCD)

Image Enhancement
(Assignment 02)

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I. Introduction

Image enhancement is one of the most important and fundamental techniques in digital image processing. This technique is used to improve the visual quality of an image. The goal of image enhancement is to make the existing information in an image clearer and easier to see. In this assignment, four different enhancement techniques are applied to four types of images, which are blurred, overexposed, dark, and low contrast images. Then, the results are analyzed to understand how each method affects the images and how effective each method is for different types of images that were used.

II. Photos Used

Four different images were used as input, with each image representing a different image problem needed to be enhanced. The first image is a photo of a cheetah that appears blurry. The second image is a street photo that looks too bright, representing an overexposed image. The third image is an interior photo where most details cannot be seen, representing a dark image. The last image shows two buildings, representing a low contrast image. All of these photos were resized to 512×512 pixels before being processed.

III. Enhancement Methods

Sharpness Enhancement - Blurred Image

The blurred image was enhanced using ImageEnhance.Sharpness with a factor of 13.0. This method makes the edges more defined and the details easier to see (making it sharper). A factor of 1.0 means that the image doesn't change, while a higher factor produces sharper images. I chose 13.0 as the value of the factor because the original image was heavily blurred and needed stronger sharpening to show the difference. The result shows sharper edges and more visible texture on the cheetah, although some blurriness can still be seen.

Brightness and Contrast Adjustment - Overexposed Image

The overexposed image was enhanced using ImageEnhance.Brightness with a factor of 0.6 and ImageEnhance.Contrast with a factor of 1.5. The decrease in the brightness factor makes the image darker, and the contrast factor makes the difference between light and dark areas more clearer. I chose 0.6 as the factor for the brightness because the original image was too bright, while 1.5 in the contrast factor makes the image look more balanced. The result shows more visible details and a more natural appearance.

Brightness Enhancement - Dark Image

The dark image was enhanced using ImageEnhance.Brightness with a factor of 6.0. This method makes the image brighter, allowing details in the dark areas to become visible. A factor of 6.0 was chosen because the original image was very dark and most of the details cannot be seen. The result makes details such as the staircase, railings, and other objects more visible, although some areas may appear too bright, and some are still too dark.

Contrast Enhancement - Low Contrast Image

The low contrast image was enhanced using ImageEnhance.Contrast with a factor of 2.0. This method increases the difference between the light and dark areas inside the image, making the details easier to see. A factor of 2.0 was chosen because the original image had only few differences between the light and dark areas. The result makes the buildings more visible and distinct compared to the original.

IV. Analysis and Comparison

Image Type	Method	Factor Value	Result
Blurred	Sharpness Enhancement	13.0	Edges more defined and details are more sharper
Overexposed	Brightness & Contrast Adjustment	0.6 & 1.5	Bright areas are reduced and details become more visible
Dark	Brightness Enhancement	6.0	Dark areas become brighter and hidden details become more visible
Low Contrast	Contrast Enhancement	2.0	Light and dark areas become more different and details become clearer

Table 4.1 Comparison of Image Enhancement Methods and its Result

V. Conclusion

Ultimately, this assignment demonstrated four image enhancement techniques on four different types of problematic images. As shown in Table 4.1, each method was used based on the problem in the image. Sharpness enhancement makes the blurred image sharper, brightness and contrast adjustments improves overexposed images, brightness enhancement makes the dark image brighter, and contrast enhancement makes the low contrast image clearer. The results show that each method can improve specific images based on the problem that occurs, with the factor value based on the severity of the problem in the image. Overall, the results show that the suitable enhancement method and factor value depend on the type and severity of the problem in the image.